

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Medium

Question

$$\frac{1}{4}(x + 5) - \frac{1}{3}(x + 5) = -7$$

What value of x is the solution to the given equation?

Answer

- A. ~~−12~~
- B. ~~−5~~
- C. **79**
- D. ~~204~~

Correct Answer: C

Rationale

Choice C is correct. For the given equation, $(x + 5)$ is a factor of both terms on the left-hand side. Therefore, the given equation can be rewritten as $(\frac{1}{4} - \frac{1}{3})(x + 5) = -7$, or $(\frac{3}{12} - \frac{4}{12})(x + 5) = -7$, which is equivalent to $-\frac{1}{12}(x + 5) = -7$. Multiplying both sides of this equation by -12 yields $x + 5 = 84$. Subtracting 5 from both sides of this equation yields $x = 79$.

Choice A is incorrect. This is the value of x for which the left-hand side of the given equation equals $\frac{7}{12}$, not -7 .

Choice B is incorrect. This is the value of x for which the left-hand side of the given equation equals 0 , not -7 .

Choice D is incorrect. This is the value of x for which the left-hand side of the given equation equals $-\frac{209}{12}$, not -7 .